

HOSPITAL MANAGEMENT SYSTEM

USE CASE FLOW

UC-01 – Patient Registration & Login

Actor: Patient

Main Flow

1. Patient opens the system.
2. Patient selects Register.
3. Patient enters name, age, sex, phone number and email.
4. System validates the entered details.
5. System sends an OTP to the registered phone number.
6. Patient enters the OTP.
7. System verifies the OTP and creates the patient account.
8. Patient enters valid login credentials.
9. System authenticates the patient and provides access.

UC-02 – OTP Verification

Actor: Patient

Main Flow

10. Patient provides the registered phone number.
11. System generates and sends an OTP.
12. Patient enters the received OTP.
13. System validates the OTP.
14. If the OTP is correct, the phone number is verified.
15. System allows the patient to continue registration/login.

UC-03 – View Medical History

Actor: Patient / Doctor

Main Flow

16. User logs into the system.
17. User selects the patient's medical history.
18. System verifies user authorization.
19. System retrieves the patient's stored medical history and reports.
20. System displays the available medical information.

UC-04 – Doctor Availability & Appointment Booking

Actor: Patient

Main Flow

21. Patient logs into the system.
22. Patient selects the required medical field.
23. System displays available doctors.
24. Patient selects a doctor.
25. System displays available appointment schedules.
26. Patient selects an available slot.
27. System confirms and stores the appointment.
28. System displays the appointment confirmation.

UC-05 – Appointment Cancellation

Actor: Patient

Main Flow

29. Patient logs into the system.
30. Patient views booked appointments.
31. Patient selects an appointment to cancel.
32. System displays the appointment details.
33. Patient confirms cancellation.
34. System updates the appointment status to cancelled.
35. System makes the slot available again.

UC-06 – Medical Diagnosis & Reports

Actor: Doctor

Main Flow

36. Doctor logs into the system.
37. Doctor selects the patient's appointment/record.
38. System displays the patient's available medical history.
39. Doctor examines the patient.
40. Doctor enters the initial diagnosis.
41. Doctor specifies the required reports/scans.
42. System saves the diagnosis and required reports/scans to the patient's record.

UC-07 – Prescription Management

Actor: Doctor / Patient

Main Flow

43. Doctor logs into the system.
44. Doctor opens the patient's medical record.
45. Doctor enters the final prescription and medication details.
46. System validates and stores the prescription.
47. Patient logs into the system.
48. Patient opens the prescription section.
49. System displays the stored prescription.

UC-08 – Online Payment & Medicine Purchase

Actor: Patient

Main Flow

50. Patient views the prescribed medicines.
51. Patient selects the required medicines.
52. System calculates the payment amount.
53. Patient selects online payment.
54. Patient provides the required payment details.
55. System processes the payment securely.
56. System confirms successful payment.
57. System records the medicine purchase.

UC-09 – File Download

Actor: Patient

Main Flow

58. Patient logs into the system.
59. Patient opens the medical documents section.
60. System displays available reports, prescriptions and other files.
61. Patient selects the required file.
62. Patient selects Download.
63. System downloads the selected file to the patient's device.

UC-10 – Check Bed Availability

Actor: Hospital Staff / Patient

Main Flow

64. User accesses the bed availability section.
65. System retrieves the current bed information.
66. System checks the status of available beds.
67. System displays the current bed availability as available or occupied.

Alternative / Exception Flows

UC-01: If the entered login credentials are invalid, the system displays an error and asks the patient to enter the credentials again.

UC-02: If the OTP is incorrect, the system rejects the verification and asks the patient to enter the correct OTP.

UC-03: If the user is not authorized, the system denies access to the patient's medical information.

UC-04: If no appointment slot is available, the system informs the patient and displays other available schedules/doctors.

UC-05: If the patient does not confirm cancellation, the appointment remains active.

UC-06: If required patient information is unavailable, the doctor can proceed with the available information and enter the diagnosis.

UC-07: If prescription details are incomplete, the system asks the doctor to provide the required information before saving.

UC-08: If payment fails, the system displays the payment failure and allows the patient to retry.

UC-09: If the selected file is unavailable, the system displays an appropriate error message.

UC-10: If no beds are available, the system displays that all required beds are currently occupied.